The end result is a visual experience of the effect of the `.layout()` function. This function specifies the number of passes that `d3.layout.sankey` makes to determine the best position of the lines representing flow. You can see some snapshots of this in figure 5.26 showing the lines sort out and get out of each other's way. This kind of position optimization is a common technique in information visualization, and drives the force-directed network layout that you'll see in chapter 6. In the case of our Sankey example, even one pass of the layout provides good positioning. That's because this is a simple dataset, and it stabilizes quickly. As you can see as you click your chart and in figure 5.26, the layout doesn't change much with progressively higher numbers of passes in the `layout()` setting.

It should be clear by this example that when you update the settings of the layout, you can also update the visual display of the layout. You can use animations and transitions by simply calling the elements and setting their drawing code or position to reflect the changed data. You'll see much more of this in later chapters.

5.6.2 Word clouds

One of the most popular information visualization charts is also one of the most maligned: the word cloud. Also known as a tag cloud, the word cloud uses text and text size to represent the importance or frequency of words. Figure 5.27 shows a



Figure 5.27 A word or tag cloud uses the size of a word to indicate its importance or frequency in a text, creating a visual summary of text. These word clouds were created by the popular online word cloud generator Wordle (www.wordle.net).